From Noise to Diversity: Random Embedding Injection in LLM Reasoning

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Abstract

Recent *soft prompt* methods, which prepend or insert trainable continuous embeddings into LLM inputs as virtual tokens, optimize these embeddings to enhance mathematical reasoning. However, the mechanism by which embedding injection itself affects model behavior, independent of optimization, remains unexplored. We investigate Random Soft Prompts (RSPs), continuous vectors sampled from the model's token embedding distribution and injected without any training. Empirically, RSPs increase early-stage token entropy while generation stabilizes thereafter, and combining RSPs with temperature sampling yields higher trajectory diversity as measured by Pass@n. Theoretically, we provide two complementary analyses: an energy-based framework showing that RSP introduces Langevin-like perturbations on the Hopfield energy landscape with natural dual decay across layers and generation steps, and a distributional analysis showing that RSP induces stochastic token rankings and expands the output support beyond temperature sampling. We further discuss the implications for GRPO training, where trajectory diversity is critical for learning signal quality.

1 Introduction

As large language models (LLMs) scale to billions of parameters, full fine-tuning becomes increasingly expensive. This has motivated parameter-efficient methods that adapt frozen models by introducing a small number of trainable parameters [Hu et al., 2022, Houlsby et al., 2019]. Among these, *soft prompts* prepend or insert learnable continuous embeddings into the input sequence, steering model behavior through the attention mechanism [Li and Liang, 2021, Lester et al., 2021]. This paradigm has been actively adopted for enhancing mathematical reasoning in LLMs [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Hao et al., 2025, Zhang et al., 2025]. Despite their diverse designs, these methods share a common structure: they inject continuous embeddings that lie outside the pretrained vocabulary and optimize them to improve downstream performance.

However, we find that optimization is not a necessary condition for these effects. We observe that injecting randomly initialized, untrained continuous embeddings can also shift the output distribution. For instance, applying a chat template to a base model not fine-tuned with such format causes a prompt-format mismatch that degrades reasoning capability; appending random embeddings to such mismatched inputs mitigates this degradation. This suggests that injecting out-of-vocabulary continuous embeddings itself, independent of any learned content, plays a non-trivial role in shaping model behavior. Existing studies primarily evaluate *soft prompt* methods through end-task accuracy, and the mechanisms by which embedding injection influences the output distribution remain insufficiently explored.

In this work, we investigate the role of Random Soft Prompts (RSPs), continuous vectors sampled from the Gaussian distribution of the model's token embedding matrix and injected without any

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training. Empirically, RSPs increase early-stage token entropy while the output stabilizes in later generation stages, and combining RSPs with temperature sampling yields more diverse reasoning trajectories as measured by Pass@n scaling. Theoretically, we analyze RSPs from two perspectives: an energy-based framework showing that RSP introduces controlled stochastic perturbations with natural decay through the attention mechanism, and a distributional analysis showing that RSP induces stochastic token rankings and expands the output support beyond what temperature sampling can achieve. We further discuss the implications of these findings for Group Relative Policy Optimization (GRPO) [Shao et al., 2024], where trajectory diversity is essential for effective learning signals.

Our contributions are as follows:

- We empirically analyze how RSPs affect LLM generation, showing that RSPs increase early token entropy and produce more diverse reasoning trajectories when combined with temperature sampling.
- We provide theoretical analyses from two perspectives: an energy-based framework explaining why RSP can perturb the output distribution without degrading performance, and a distributional analysis showing that RSP induces stochastic token rankings and expands the output support beyond the reach of temperature sampling.
- We discuss the implications of RSP-induced diversity for GRPO training and argue that RSP provides exploration gains complementary to existing temperature-based approaches.

2 Background

2.1 Soft Prompt Methods for LLM Reasoning

Soft prompts emerged as a parameter-efficient alternative to full fine-tuning. Prefix-Tuning [Li and Liang, 2021] introduced the paradigm by prepending trainable continuous vectors to every transformer layer, Prompt Tuning [Lester et al., 2021] simplified this to input-layer-only tuning and showed that it matches full fine-tuning at sufficient scale, and P-tuning [Liu et al., 2023] extended the approach with an LSTM-based prompt encoder to model inter-position dependencies.

More recently, *soft prompts* have been actively applied to LLM reasoning. TTSV [Kang et al., 2026] optimizes prefix embeddings at test time via entropy minimization, steering the model toward high-certainty states to activate latent reasoning capabilities. SoftCoT [Xu et al., 2025] replaces explicit chain-of-thought tokens with soft tokens generated by an assistant model, leveraging the stronger representational capacity of continuous representations over discrete language. LTPO [Ye et al., 2026] optimizes latent thought embeddings per test instance through policy gradient with a confidence-based intrinsic reward. COCONUT [Hao et al., 2025] feeds the model's own hidden states back as input embeddings, enabling reasoning in a continuous latent space. MemGen [Zhang et al., 2025] demonstrates that embedding vectors can serve as experience memory by constructing latent token sequences that enrich the model's reasoning. Pause tokens [Goyal et al., 2024] insert learnable tokens into the input to provide additional computation steps.

These approaches all inject out-of-vocabulary continuous embeddings optimized for task-specific objectives. Because evaluation centers on end-task accuracy, the contribution of the injection itself and that of the optimization are not separated, and the mechanisms by which such out-of-vocabulary continuous embeddings shape model behavior remain a separate and underexplored question.

2.2 Noise Injection in Neural Networks

Neural networks have incorporated noise at various stages. During training, dropout [Srivastava et al., 2014] randomly deactivates hidden units to prevent overfitting, and NEFTune [Jain et al., 2024] adds noise to token embeddings during instruction fine-tuning, reporting improvements in instruction following. These methods use noise as a regularizer and do not apply perturbations at inference time.

At inference time, perturbation-based approaches primarily target robustness and safety. Randomized smoothing [Cohen et al., 2019] injects Gaussian noise into inputs to obtain certified robustness guarantees. In the LLM setting, SmoothLLM [Robey et al., 2023] perturbs input prompts at the character level to defend against adversarial jailbreaking. These methods require multiple noisy forward passes with output aggregation, and target prediction stability rather than output diversity.

Table 1: Accuracy (%) across model configurations and benchmarks. RSP reports the best result across injection positions (Prefix, Infix, Suffix); full per-position breakdown is in Appendix A. Values in parentheses indicate the difference from the baseline.

	MATH-500		GSM8K		AIME24	
Model	Baseline	RSP	Baseline	RSP	Baseline	RSP
Instruct Models						
LLaMA-3.1-8B-Inst	52.20	49.40 (−2.8)	85.75	86.73 (+1.0)	6.67	10.00 (+3.3)
Qwen2.5-Math-7B-Inst	83.20	84.20 (+1.0)	95.45	95.68 (+0.2)	13.33	16.67 (+3.3)
Qwen2.5-Math-1.5B-Inst	73.00	75.20 (+2.2)	85.29	85.75 (+0.5)	10.00	20.00 (+10.0)
Base Models (Raw Text)						
Qwen2.5-Math-7B	72.20	71.60 (−0.6)	84.15	84.31 (+0.2)	6.67	16.67 (+10.0)
Qwen2.5-Math-1.5B	61.40	64.40 (+3.0)	77.86	74.30 (−3.6)	16.67	10.00 (−6.7)
Base Models + ChatML (Format Mismatch)						
Qwen2.5-Math-7B	52.20	70.40 (+18.2)	58.30	75.97 (+17.7)	23.33	23.33 (+0.0)
Qwen2.5-Math-1.5B	34.40	63.40 (+29.0)	37.45	67.02 (+29.6)	13.33	20.00 (+6.7)

In reinforcement learning, NoisyNet [Fortunato et al., 2018] injects learnable noise into network weights to improve exploration, demonstrating that the location and structure of noise injection affect exploration quality. This motivation is closest to RSPs, which similarly target diversity, but differ in injection target (input embeddings vs. network weights) and require no learned noise parameters.

RSPs differ from these approaches in the following respects. **(1)** RSPs operate solely at inference without any training, whereas dropout and NEFTune inject noise during training, modifying the resulting model parameters accordingly. **(2)** RSPs preserve the original input and append fresh embedding vectors in a single forward pass, whereas robustness methods corrupt the input and aggregate over multiple noisy passes. **(3)** RSPs use purely random vectors sampled from the pretrained embedding statistics, whereas NoisyNet learns noise parameters through optimization.

3 Experimental Analysis

3.1 Random Soft Prompt: Definition and Setup

Let $\mathbf{W}_E \in \mathbb{R}^{V \times d}$ denote the token embedding matrix of a pretrained LLM, where V is the vocabulary size and d is the hidden dimension. We define an RSP $\mathbf{H} = [h_1, \ldots, h_L] \in \mathbb{R}^{L \times d}$ as a sequence of L continuous vectors, where each vector is independently sampled from:

$$h_j \sim \mathcal{N}(\mu_E,\ \sigma_E^2 \mathbf{I}), \quad j = 1, \ldots, L, \tag{1}$$

with $\mu_E = \frac{1}{Vd} \sum_{i,k} [\mathbf{W}_E]_{i,k}$ and $\sigma_E = \mathrm{std}(\mathbf{W}_E)$. Let $\mathbf{X} = [x_1, \ldots, x_T] \in \mathbb{R}^{T \times d}$ denote the embedding sequence of the original input. The RSP-augmented input $\tilde{\mathbf{X}}$ is constructed by concatenating $\mathbf{H}$ with $\mathbf{X}$ at one of three positions without modifying the original prompt text: *prefix* ($[\mathbf{H}; \mathbf{X}]$), *infix* ($\mathbf{H}$ inserted within $\mathbf{X}$), or *suffix* ($[\mathbf{X}; \mathbf{H}]$), following the injection strategies adopted in prior work [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Zhang et al., 2025]. Sampling a new $\mathbf{H}$ for each batch ensures that the results are not dependent on any realization of the random embeddings.

We evaluate on three mathematical reasoning benchmarks: MATH-500 [Lightman et al., 2024], GSM8K [Cobbe et al., 2021], and AIME24, using LLaMA-3.1-8B-Instruct [Grattafiori et al., 2024] and the Qwen2.5-Math model family [Yang et al., 2024] across seven configurations including instruct models, base models, and base models with mismatched chat templates. All experiments use greedy decoding to isolate the effect of RSPs from sampling stochasticity. Only the final answer is evaluated through a fallback-based extraction pipeline. Full experimental details are provided in Appendix A.

3.2 Does RSP Degrade Reasoning Performance?

We first examine how injecting random embeddings affects the model's reasoning performance. Table 1 reports the accuracy for each model configuration, using the best RSP result across injection positions (Prefix, Infix, Suffix) and $L \in \{10, 15, 20\}$ for each benchmark.

Table 2: Comparison with existing soft prompt methods (TTSV, SoftCoT, LTPO) under unified evaluation. Baseline and RSP values are from Table 1. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Benchmark	Baseline	RSP	TTSV	SoftCoT	LTPO
Qwen2.5-Math-7B-Instruct	MATH-500	83.20	**84.20**	83.80	82.80	83.00
	GSM8K	95.45	**95.68**	95.30	95.00	94.84
	AIME24	13.33	16.67	**23.30**	16.67	13.33
Qwen2.5-Math-1.5B-Instruct	MATH-500	73.00	75.20	75.20	**76.00**	72.40
	GSM8K	85.29	**85.75**	85.70	84.46	84.53
	AIME24	10.00	**20.00**	6.70	6.67	10.00

For instruct models and base models with their native prompt formats, RSP injection maintains or slightly shifts baseline performance within a few percentage points. Most strikingly, under prompt-format mismatch — where base models receive ChatML templates they were not trained on — RSP yields pronounced improvements of up to +29%p. These gains are difficult to attribute to simple perturbation effects, as noise that merely raises model uncertainty would further degrade performance.

We compare RSPs with prior soft prompt methods (TTSV, SoftCoT, LTPO) that optimize their embeddings for distinct objectives, under a unified evaluation protocol reported in Table 2. All methods are re-evaluated with a shared answer extraction pipeline that applies fallback strategies, enabling a format-agnostic comparison despite each method's distinct prompt format and grading scheme. The results place RSPs on par with the optimized methods without requiring any training.

3.3 How Does RSP Affect the Output Distribution?

We analyze how RSP injection changes the model's output distribution during generation on Qwen2.5-Math-1.5B-Instruct with MATH-500, measuring per-token entropy, top-1 probability, and varentropy across the generation process. Figure 1 and Table 3 compare these metrics for the first 5% of generation steps across RSP variants and prior soft prompt methods.

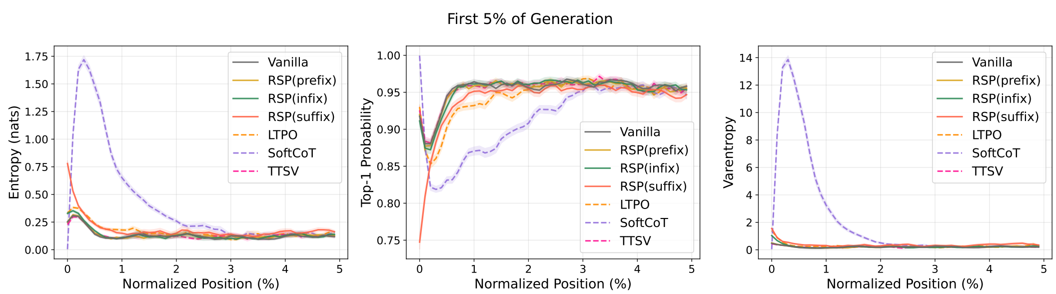

Figure 1: Mean entropy, top-1 probability, and varentropy during the first 5% of generation steps (Qwen2.5-Math-1.5B-Instruct, MATH-500). Shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: existing soft prompt methods (LTPO, SoftCoT, TTSV).

Table 3: Mean entropy, top-1 probability, and varentropy for the first 5% of generation steps (Qwen2.5-Math-1.5B-Instruct, MATH-500).

Metric	Baseline	Prefix	Infix	Suffix	LTPO	SoftCoT	TTSV
Entropy	0.1332	0.1366	0.1402	0.1941	0.1662	0.4218	0.1359
Top-1 Prob	0.9535	0.9523	0.9525	0.9373	0.9437	0.9156	0.9534
Varentropy	0.2181	0.2207	0.2463	0.4010	0.2953	2.2234	0.2174

These results show that RSP modifies the early-stage output distribution across all injection positions: entropy increases (the token distribution flattens), top-1 probability decreases (commitment to a dominant token weakens), and varentropy increases (indicating multimodal distributions [Ahmed et al., 2026]). High-entropy positions are known to act as critical forking points that determine reasoning trajectories [Wang et al., 2025], so this shift moves the model toward a state where multiple reasoning paths can coexist. The change is localized to the initial reasoning stage; in the middle and final portions of generation, all three metrics converge to baseline levels.

Although existing soft prompt methods are optimized for different objectives, they share a common pattern of increasing early-stage entropy. Notably, TTSV, which optimizes soft prompts using a reward that reduces the mean entropy of reasoning trajectories, still exhibits early entropy comparable to the baseline. This suggests that elevating early-stage entropy is an inherent effect of injecting non-pretrained continuous embeddings, independent of the optimization objective.

3.4 Does RSP Induce Trajectory Diversity?

Section 3.3 empirically established that RSP reshapes the early-stage output distribution. We now examine whether this shift translates into reasoning diversity at three complementary levels of analysis: token rankings, semantic content of trajectories, and task-level outcomes. All three analyses share the setting: Qwen2.5-Math-1.5B-Instruct, MATH-500, suffix injection, and $L = 10$.

Token-level: distribution beyond temperature. Under autoregressive decoding, any trajectory-level divergence between RSP and the baseline must originate from the first generation step, a critical forking point for reasoning trajectories [Wang et al., 2025]. Using the baseline at τ=1.0 as reference, we compare RSP at τ=1.0 against the baseline at τ=2.0 (the high-temperature ceiling of what flattening alone can achieve) on three complementary metrics: Spearman rank correlation ρ to detect rank reorganization, the probability mass placed outside the baseline's top-10 to measure support expansion, and Jensen–Shannon divergence to quantify the overall magnitude of distributional change (formal definitions in Appendix C).

Table 4: First-token distribution metrics measured against the baseline at τ=1.0, comparing the baseline at τ=2.0 with RSP at τ=1.0.

Metric	τ=2.0	**RSP**
Spearman ρ	1.0	**0.651**
Mass outside top-10	5.15%	**27.64%**
JS divergence	0.086	**0.231**

Table 4 reports the contrast. Even at τ=2.0, only 5.15% of probability mass falls outside the baseline's top-10; RSP at τ=1.0 reaches 27.64%. The Spearman rank correlation tells the story more starkly: because temperature scaling $p_i^{(\tau)} \propto p_i^{1/\tau}$ is a monotone transform that preserves token ranking ($p_i > p_j \iff p_i^{(\tau)} > p_j^{(\tau)}$) for any $\tau > 0$, ρ relative to the baseline is exactly 1 as a mathematical identity. RSP, by contrast, reranks the token distribution itself, reducing ρ to 0.651—an effect temperature cannot produce at any τ. The Jensen–Shannon divergence (0.231) quantifies the overall magnitude of this distributional change. Since $\rho < 1$ rules out rank-preserving flattening as the sole cause, the high divergence reflects a structural reshape of the distribution rather than a uniform flattening of the same ranking.

Trajectory-level: semantic diversity. Token-level reranking raises a second question: do first-token perturbations produce semantically distinct reasoning paths, or do independently perturbed trajectories converge to the same solution strategy? We randomly sample 64 problems from MATH-500 and generate 300 independent trajectories per problem under two conditions, Baseline and RSP, with temperature $\tau = 1.0$. The trajectories are embedded via BGE-M3 [Chen et al., 2024] and compared on three metrics: mean pairwise cosine distance within each problem, inter-cluster distance between DBSCAN [Ester et al., 1996] centroids, and intra-cluster distance. Table 5 shows that RSP substantially increases the inter-cluster distance while leaving the intra-cluster distance unchanged: clusters become more separated while reasoning within each cluster remains consistent. RSP also wins on pairwise distance in 56 of 64 problems (87.5%).

Table 5: Semantic diversity metrics averaged across the 64 problems.

Metric	**Baseline**	**RSP**
Pairwise cos. dist.	0.0612	**0.0879**
Inter-cluster dist.	0.0183	**0.0982**
Intra-cluster dist.	0.0526	0.0528

Figure 2(a) illustrates the qualitative difference for the representative problem: the RSP samples (blue) cover the region occupied by baseline samples (red) while extending into clusters that the baseline does not reach; the expanded regions also contain correct solutions (filled markers), showing that RSP's exploration is not limited to incorrect alternatives but reaches new reasoning paths that still yield the correct answer. Figures 2(b) and 2(c) show that this pattern holds across the 64 problems: on most problems RSP produces more clusters and larger pairwise cosine distance than baseline. The combination of large inter-cluster increase and unchanged intra-cluster distance mirrors the token-

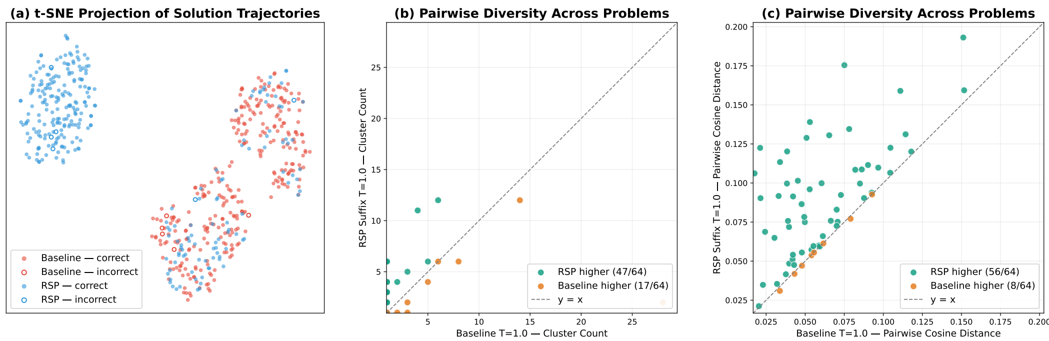

Figure 2: Semantic diversity analysis on 64 randomly sampled MATH-500 problems with 300 trajectories per condition (Qwen2.5-Math-1.5B-Instruct, $\tau = 1.0$, BGE-M3 embeddings). (a) t-SNE projection for one representative problem (`test/prealgebra/951`): baseline (red) concentrates around one region while RSP suffix (blue) spreads across distinct clusters; filled markers denote correct answers and open markers denote incorrect. (b) Per-problem mean pairwise cosine distance; points above the diagonal indicate RSP produces more diverse trajectories than baseline. (c) Per-problem DBSCAN cluster count; RSP yields more distinct clusters than baseline on most problems.

level finding: RSP alters which cluster a trajectory enters while preserving the local continuation within each cluster. Additional per-problem t-SNE plots are provided in Figure 6 (Appendix B).

Outcome-level: Pass@n scaling. Finally, we evaluate whether the token- and trajectory-level diversity translates into task-level outcome diversity through Pass@n, the probability that at least one of n sampled solutions is correct. We compare four conditions with $n = 16$ samples per problem on MATH-500 and AIME24: *(i) Baseline,* $\tau = 1.0$: temperature sampling only; *(ii) RSP (fixed seed),* $\tau = 1.0$: a single RSP shared across all samples; *(iii) RSP (indep. seed), greedy*: a different RSP per sample without temperature; and *(iv) RSP (indep. seed),* $\tau = 1.0$: a different RSP per sample combined with temperature sampling.

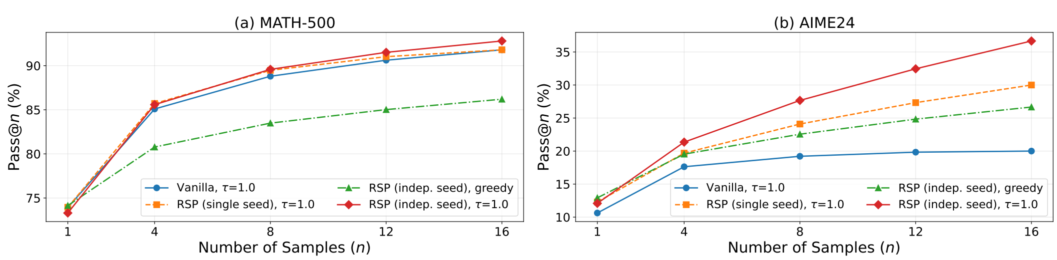

Figure 3: Pass@n scaling on MATH-500 (left) and AIME24 (right) with Qwen2.5-Math-1.5B-Instruct, $n = 16$ samples per problem. Baseline: temperature sampling only. Fixed seed: single RSP shared across samples combined with temperature. Indep. seed: a different RSP per sample, with or without temperature.

Condition *(iv)* achieves the highest Pass@n on both benchmarks, reaching 92.80% on MATH-500 and 36.67% on AIME24 at $n = 16$. Condition *(ii)* shows no improvement over the baseline, confirming that the diversity gain originates from varying the random embeddings across samples, not from a single fixed perturbation. Pass@1 remains comparable to the baseline on MATH-500, and on the more challenging AIME24, *(iv)* achieves higher Pass@1 than the baseline. Independent RSPs combined with temperature sampling produce the most diverse reasoning trajectories while maintaining or improving single-sample accuracy.

Synthesis across the three levels. The three analyses converge on a consistent picture. At the token level, RSP induces support expansion concentrated at the first token and decaying monotonically across steps. At the trajectory level, this localized perturbation translates into distinct reasoning strategies (large inter-cluster separation) while preserving local coherence within each strategy (unchanged intra-cluster distance). At the outcome level, the resulting trajectories cover more correct solutions without sacrificing single-sample accuracy.

4 Theoretical Analysis: Why RSP Works

Our theoretical goal is narrower than a theorem about the full nonlinear transformer. We want a mechanism that explains the three empirical regularities from Section 3: random continuous prompts can alter the model's earliest reasoning decisions even though they are not optimized for the task; their influence is strongest at the beginning of decoding and weakens later; and this early perturbation can increase trajectory diversity without permanently destabilizing generation. The common control variable throughout the section is $w_{r,i}^{(l)}$, the attention mass assigned to the random tokens. The argument therefore has three parts. First, we characterize what kind of perturbation can open reasoning branches without inserting a hand-crafted prior, and explain why centered isotropic randomness is the relevant choice. Second, we show that attention injects the perturbation through $w_{r,i}^{(l)}$ and simultaneously places an annealing pressure on it as real-token evidence grows. Third, we show why this pre-softmax perturbation can change token rankings in a way that temperature scaling cannot.

4.1 Why Random Rather Than a Fixed Token?

A generic exploration prompt for reasoning should satisfy three structural requirements. It should not systematically pre-commit the model to one semantic direction before the input-dependent computation begins. It should vary across trials, otherwise repeated decoding would trace the same perturbation every time. And because the branch-sensitive direction is unknown in advance and depends on the prompt, the decoding state, and the token comparison, the perturbation should not privilege some directions while leaving others almost untouched. Optimized soft prompts deliberately encode a task prior [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Zhang et al., 2025]; our question is the complementary one: what perturbation should one use when no such prior is assumed?

A fixed token or a fixed continuous prompt fails this requirement set immediately. It may steer one trajectory, but if the same vector is reused across runs then there is no across-trial exploration at all. Resampling discrete vocabulary tokens restores variability, yet each draw still comes with its own lexical direction and there is no reason for the resulting perturbations to be centered or isotropic. We formalize the choice we make instead.

Definition 1 (Random Soft Prompt)**.** *Let $\mu_E \in \mathbb{R}^d$ and $\sigma_E^2 \in \mathbb{R}_{>0}$ denote the per-coordinate mean and variance of the model's input-embedding distribution. A* Random Soft Prompt *(RSP) of length L is a sequence $\mathbf{H} = (h_1, \ldots, h_L)$ with $h_l \overset{iid}{\sim} \mathcal{N}(\mu_E, \sigma_E^2 \mathbf{I}_d)$, drawn independently per inference call. Writing $\bar{h}_l := h_l - \mu_E$, the centered prompt $\bar{h}_l \sim \mathcal{N}(0, \sigma_E^2 \mathbf{I}_d)$ is the formal object analyzed in this section.*

The two design choices in Definition 1—independent resampling and centering—together produce a centered, isotropic, finite-variance perturbation in prompt space, which is what the lemma and theorem below require.

Lemma 1 (Semantic neutrality)**.** *Let $\bar{h} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_d)$ and $\hat{h} := \bar{h}/\|\bar{h}\|$. For any fixed unit vectors $u_1, \ldots, u_M \in \mathbb{S}^{d-1}$ and any $\delta \in (0, 1)$,*

$$\Pr\left[\max_{r \in [M]} |\hat{h}^\top u_r| \leq \sqrt{\frac{2\log(2M/\delta)}{d-1}}\right] \geq 1 - \delta.$$

In contrast, for any nonzero token embedding e_s, choosing the anchor direction $u = \hat{e}_s := e_s/\|e_s\|$ yields $|\hat{e}_s^\top u| = 1$.

Lemma 1 gives the precise sense in which the perturbation is neutral. The claim is not that a random vector is literally free of every incidental semantic correlation. The relevant and weaker statement is that it does not systematically align with any pre-selected finite family of semantic alternatives. Centering then removes deterministic first-order drift: under the local surrogate below, the expected perturbation of any logit gap is zero when $\mathbb{E}[\bar{h}] = 0$.

To formalize the exploration part, let $\bar{h}$ be the centered prompt vector and consider the local first-order surrogate

$$z_i(\bar{h}) \approx z_i + a_i^\top \bar{h}, \qquad a_i := \nabla_{\bar{h}} z_i(0),$$

valid for sufficiently small $\|\bar{h}\|$ (Appendix E.1); we use this surrogate throughout the section to identify the direction of perturbation rather than to predict its exact magnitude. For a token pair (i, j), the perturbed logit gap becomes

$$\Delta_{ij}(\bar{h}) \approx \Delta_{ij} + b_{ij}^\top \bar{h}, \qquad b_{ij} := a_i - a_j.$$

The relevant question is therefore not "does the perturbation have variance?" but "how much variance does it retain along the unknown direction b_{ij} that matters for this branching decision?" The theorem below answers that question at the level of second moments.

Theorem 1 (Direction-agnostic coverage)**.** *Let D be any zero-mean perturbation distribution on $\mathbb{R}^d$ with covariance Σ_D and variance budget $\operatorname{tr}(\Sigma_D) \le \rho^2$. Define the worst directional variance*

$$\mathcal{E}(D) := \min_{\|b\|=1} \operatorname{Var}_{h\sim D}(b^\top h).$$

Then

$$\mathcal{E}(D) = \lambda_{\min}(\Sigma_D) \le \frac{\rho^2}{d},$$

with equality if and only if $\Sigma_D = (\rho^2/d)\mathbf{I}_d$. Hence isotropic covariance is the unique way to equalize first-order perturbation variance across all directions while maximizing the least-covered one.

Theorem 1 is not meant to cast prompting as an adversarial game; its role is simpler. When the branch-sensitive direction is unknown at sample time, isotropy is the unique second-order description that neither favors nor neglects any direction. A fixed prompt has $\Sigma_D = 0$ and therefore creates no across-trial exploration at all. An anisotropic perturbation leaves at least one direction less explored than the isotropic optimum. The Gaussian choice in RSP is the simplest centered, rotation-symmetric family that realizes this covariance while matching the scale of pretrained embeddings. This is the sense in which RSP is not just "some noise": it is a resampled, centered, direction-agnostic perturbation.

A natural alternative one might consider is to draw the perturbation from a distribution that *matches* the empirical geometry of the pretrained embedding table $\mathbf{W}_E$, rather than discarding it. Theorem 1 dismisses this alternative as soon as the embedding geometry is known to be anisotropic.

Corollary 1 (Embedding-matched noise is suboptimal in worst-direction coverage)**.** *Let Σ_E be the empirical covariance of the rows of $\mathbf{W}_E$, and let $\rho^2 := \operatorname{tr}(\Sigma_E)$. Consider two zero-mean perturbations on $\mathbb{R}^d$ with the same trace budget ρ^2: the isotropic RSP law D_{iso} with covariance $(\rho^2/d)\mathbf{I}_d$, and the embedding-matched law D_{emb} with covariance Σ_E. Then*

$$\mathcal{E}(D_{\text{iso}}) \;=\; \frac{\rho^2}{d} \;\ge\; \lambda_{\min}(\Sigma_E) \;=\; \mathcal{E}(D_{\text{emb}}),$$

with strict inequality whenever Σ_E is non-isotropic.

The corollary is one application of Theorem 1 and requires no separate proof. Its empirical bite comes from the fact that pretrained LLM token-embedding matrices are documented to be highly anisotropic, concentrating in a low-dimensional "cone" inside $\mathbb{R}^d$ [Mu and Viswanath, 2018, Ethayarajh, 2019]. In that regime $\lambda_{\min}(\Sigma_E)$ is orders of magnitude smaller than ρ^2/d, so the worst-covered direction under embedding-matched noise is nearly unexplored—precisely the directions a generic exploration prompt should not abandon. Discarding the directional geometry of $\mathbf{W}_E$ while keeping its scale is therefore not a simplification of an embedding-matched alternative but a quantitatively better choice for direction-agnostic exploration.

Lemma 1 and Theorem 1 therefore play complementary roles. The lemma rules out semantic pre-commitment to any fixed anchor set, while the theorem rules out directional under-coverage in the local branch-sensitive geometry. A perturbation that satisfies only the first condition would be neutral but too weak in some directions; a perturbation that satisfies only the second would still inject a hand-chosen semantic prior. RSP is useful because it satisfies both at once.

4.2 How Attention Turns a Neutral Perturbation into Controlled Exploration

The previous subsection explains why centered isotropic randomness is the right kind of perturbation. The next question is temporal: why is the perturbation strong enough to matter at the beginning of decoding but weak enough not to derail the rest of the reasoning chain? The formal answer needs only two ingredients. First, the attention output admits an exact decomposition into a real-token update and a random-token contribution. Second, the same attention mass that injects the random contribution is structurally forced downward as real-token evidence accumulates.

Exact decomposition. When k random vectors are inserted as additional keys and values, the attention update can be written as

$$\mathbf{x}_i^{(l+1)} = (1 - w_{r,i}^{(l)})\,\tilde{\mathbf{x}}_i^{(l+1)} + \boldsymbol{\eta}_i^{(l)}, \tag{2}$$

where $\tilde{\mathbf{x}}_i^{(l+1)}$ is the update produced by the real tokens alone, $\boldsymbol{\eta}_i^{(l)} := \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)}$ is the contribution of the random values, and

$$w_{r,i}^{(l)} := \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)}$$

is the total attention mass assigned to the random tokens. Appendix E.1 proves the decomposition and the deterministic bound

$$\|\boldsymbol{\eta}_i^{(l)}\| \le w_{r,i}^{(l)} \max_{j\in[k]} \|\mathbf{v}_{r_j}^{(l)}\|.$$

This is the structural core of the mechanism: the same scalar $w_{r,i}^{(l)}$ controls how much of the original update is retained and how large the random contribution can be. If $w_{r,i}^{(l)}$ is large, the perturbation can materially change the update; if $w_{r,i}^{(l)}$ is small, the computation collapses back toward the real-token update.

Natural annealing through attention. The second structural fact is that $w_{r,i}^{(l)}$ is not arbitrary. It satisfies

$$w_{r,i}^{(l)} \le \frac{k}{k + n \cdot \exp(\Delta_i^{(l)}/\sqrt{d})}, \tag{3}$$

where $\Delta_i^{(l)} := \min_{j\in[n]} \mathbf{q}_i^\top \mathbf{k}_j - \max_{j'\in[k]} \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}$ is the real-vs-random logit gap and n is the number of real tokens. This bound yields a clean *sequence-wise dilution effect*: for a fixed gap, every newly generated real token tightens the maximum random-token attention compatible with that state. A similar *layer-wise* decay is expected whenever deeper layers enlarge the real-vs-random gap $\Delta_i^{(l)}$. We intentionally treat that layer-wise statement as a model-side hypothesis rather than as a universal theorem, because it depends on how the model sharpens real-token retrieval with depth. The resulting prediction is simple: the perturbation should be strongest exactly when the model has the least context and many continuations remain open, and it should weaken as evidence accumulates.

Theorem 2 (Sequence-wise annealing under a fixed logit gap)**.** *Fix k and $\Delta_i^{(l)}$. The upper bound in Eq.* (3) *is strictly decreasing in the number n of real tokens. Consequently, cache growth alone imposes a monotone decrease on the largest random-token attention compatible with that logit gap.*

Theorem 2 proves a structural dilution effect, not monotone decay of the realized $w_{r,i}^{(l)}$ on every decoding path, because $\Delta_i^{(l)}$ also evolves with state. The layer-wise half is likewise model-specific and must be checked empirically.

Empirical verification. We test exactly these model-dependent steps on Qwen2.5-Math-7B under suffix injection. If $w_{r,i}^{(l)}$ really acts as an annealed exploration coefficient, then the realized attention to RSP tokens should decay across reasoning positions and layers, whereas attention to question tokens should not exhibit the same monotone pattern because they remain task-relevant throughout decoding. Figure 4 shows precisely this contrast. The figure therefore serves a specific logical role: it validates the model-side sharpening behavior that Eq. (3) by itself does not prove.

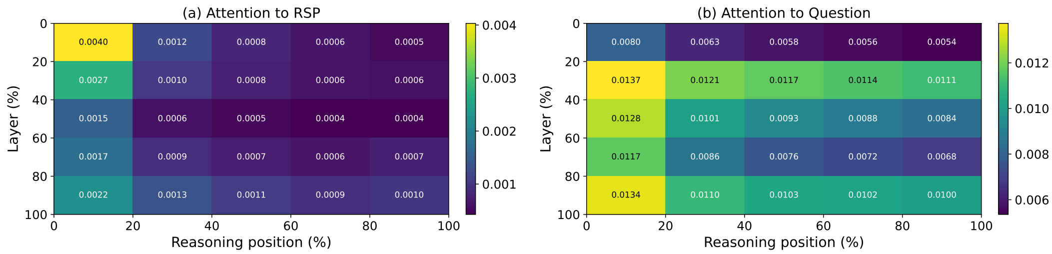

Figure 4: Per-token attention mass on Qwen2.5-Math-7B (suffix, 500 MATH-500 problems, each with an independently sampled RSP). The layer and reasoning-position axes are each partitioned into five quantile bins; every cell reports the 500-sample mean per-token attention mass directed to **(a)** the RSP embeddings and **(b)** the question tokens within that region. The horizontal axis spans reasoning position from early (left) to late (right) generation steps, and the vertical axis spans layer depth from shallow (top) to deep (bottom). Attention to the RSP embeddings in (a) decays along both axes, while attention to the question tokens in (b) remains comparatively uniform.

4.3 Output Distribution Shift and Exploration

The practical consequence of the previous mechanism is branch opening. To connect that mechanism to observable decoding behavior, we now return to the output distribution. The relevant question is not whether RSP merely flattens probabilities, but whether it can make the decoder consider tokens that temperature scaling can never promote into contention.

Why temperature is insufficient. Temperature scaling transforms probabilities as $p_i^{(\tau)} \propto p_i^{1/\tau}$ and preserves token ranking, i.e., $p_i > p_j \iff p_i^{(\tau)} > p_j^{(\tau)}$. It can soften or sharpen confidence, but it cannot create a new ordering among candidate tokens at a fixed decoding state.

Why RSP changes rankings. RSP acts before softmax, through the hidden state. At a fixed decoding state, temperature rescales existing logit gaps, whereas RSP perturbs the gaps themselves. Using the same first-order local surrogate as above, we write $z_i(\bar{h}) \approx z_i + a_i^\top \bar{h}$ for the centered prompt $\bar{h}$. The next two propositions should therefore be read as statements about this local surrogate, which is sufficient for the comparison to temperature because temperature is rank-preserving regardless of approximation.

Proposition 1 (Stochastic ranking)**.** *If there exists a token pair (i, j) such that $a_i \neq a_j$, then*

$$0 < \Pr_{\bar{h}} \big(z_i(\bar{h}) > z_j(\bar{h}) \big) < 1.$$

Thus, unlike temperature scaling, RSP randomizes token rankings at each decoding step.

Proposition 1 is the minimal statement: pairwise order can flip. Exploration requires a stronger conclusion, namely that tokens outside the baseline candidate set can actually enter it.

Proposition 2 (Top-K entry under a nonempty entry region)**.** *Let $J_K(s_t)$ be the baseline top-K set at decoding state s_t, and let $i \notin J_K(s_t)$. Define*

$$\mathcal{G}_{i,K} := \left\{ \bar{h} \in \mathbb{R}^d : \# \left\{ j \neq i : z_j + a_j^\top \bar{h} > z_i + a_i^\top \bar{h} \right\} \leq K - 1 \right\}.$$

If $\mathcal{G}_{i,K}$ contains a nonempty open set and $\bar{h}$ is sampled from an isotropic Gaussian, then

$$\Pr_{\bar{h}} \big(i \in \mathrm{TopK}(p^{\mathrm{RSP}}(\cdot \mid s_t, \bar{h})) \big) > 0.$$

Proposition 2 is intentionally conditional. It does not claim that every excluded token admits such a top-K entry region. It states that whenever the first-order model contains an open region in prompt space where a previously excluded token moves into the top-K, isotropic perturbations reach that region with positive probability. Temperature cannot do this because it never changes token order at all. The mechanism is therefore branch opening, not mere probability flattening.

From rank flips to trajectory diversity to accuracy gain. Together, Propositions 1 and 2 close the logical chain from Section 3.3 to Section 3.4, up to the local-surrogate assumption above. Early RSP perturbations alter logit gaps before softmax, which can admit previously excluded tokens into the candidate set. Because autoregressive decoding compounds early choices, one such early admission is enough to push the trajectory into a different reasoning cluster even if later perturbations are already small. RSP therefore does not need to perturb the entire chain of thought uniformly: it only needs to alter the first few branching decisions before attention-mediated attenuation suppresses further noise, which is why early exploration and late stabilization coexist in the same mechanism.

The connection from trajectory diversity to accuracy gain rests on a property of reasoning tasks rather than of RSP itself. Mathematical reasoning admits multiple valid solution paths, and a deterministically decoded trajectory commits to a single sequence of early choices that may or may not lead to a correct answer. Broader sampling over the trajectory space therefore increases the probability of reaching at least one correct chain, without requiring any particular sample to be more likely correct than the baseline. By independence of samples, N independent RSP draws admit a previously excluded token at a given decoding step with probability at least $1 - (1 - \mu_i)^N \geq 1 - \exp(-N\mu_i)$, where $\mu_i > 0$ is the per-sample admission probability guaranteed by Proposition 2. The same logic at the trajectory level explains why Section 3.4 reports both improvements and regressions across individual RSP-induced trajectories, with net accuracy gain emerging only under repeated sampling. The mechanism is exploration, not steering toward correctness, and the empirical accuracy gain is the consequence of that exploration meeting the multi-path structure of reasoning.

Taken together, the causal chain is now explicit. Randomness matters because RSP is resampled, centered, and direction-agnostic; attention turns that perturbation into an early exploration coefficient and then anneals it; the resulting early rank flips open trajectories that temperature scaling alone cannot reach; and the multi-path structure of reasoning converts trajectory diversity into accuracy gain under repeated sampling. Section 3 therefore reports four observable faces of a single mechanism rather than four disconnected phenomena.

5 Conclusion

Summary. This work investigated the role of Random Soft Prompts (RSPs), continuous embeddings sampled from the pretrained token embedding distribution and injected into the model input without any training. Through empirical and theoretical analyses, we established three findings. First, RSPs increase early-stage token entropy, top-1 probability reduction, and varentropy while the model's output stabilizes in later generation stages (Section 3). Second, RSPs induce stochastic token rankings and expand the effective support of the output distribution, a property that temperature sampling fundamentally lacks due to its ranking-preserving nature (Section 4). Third, combining RSPs with temperature sampling produces more diverse reasoning trajectories than either method alone, as measured by Pass@n scaling (Section 3).

Implications for GRPO training. GRPO computes advantages by comparing multiple sampled trajectories within each group, making trajectory diversity essential for effective learning signals. Existing approaches such as DAPO [Yu et al., 2025] and CE-GPPO [Su et al., 2025] address entropy collapse by modifying the clipping range or reintroducing gradients from clipped tokens, but the underlying sampling process remains based on temperature sampling, which preserves token rankings and cannot expand the output support. Our theoretical and empirical results suggest that RSP can address this limitation at the sampling level: by perturbing the initial output distribution with independent random embeddings per rollout, RSP provides a complementary source of diversity that is orthogonal to temperature-based exploration. We expect this to yield more effective learning signals for GRPO training, particularly in mitigating entropy collapse.

Limitations. Our experiments are conducted on mathematical reasoning benchmarks (MATH-500, GSM8K, AIME24) with a limited set of model families (Qwen2.5-Math, LLaMA-3.1). The energy-theoretic analysis in Section 4 relies on strong assumptions: the Hopfield–Attention equivalence (A1: $\mathbf{K} = \mathbf{V}$) and weight-tied layers (A2), which hold exactly only in Hopfield-type models and deep equilibrium models, respectively. Since real transformer architectures vary widely in their attention mechanisms and layer structures, the theoretical predictions are most directly applicable to models whose attention dynamics closely approximate the Hopfield energy framework. The GRPO application remains a theoretical expectation and has not been empirically validated.

Future work. Extending the experimental evaluation to other domains (e.g., code generation, commonsense reasoning) and larger model scales would test the generality of RSP's effects. Empirically validating the GRPO application by training with RSP-augmented rollouts and comparing against DAPO and CE-GPPO is a natural next step. Investigating the optimal RSP distribution and token count, which currently require per-model tuning, could lead to more principled injection strategies.

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A Experimental Setup and Full Accuracy Breakdown

Table 6 provides the full per-position accuracy breakdown (Prefix, Infix, Suffix) corresponding to the best-across-positions results summarized in Table 1 in the main text. Table 7 lists the number of RSP tokens L used for each model–benchmark pair, selected from $\{10, 15, 20\}$. All experiments are conducted on a single NVIDIA A6000 40GB GPU with greedy decoding, a maximum generation length of 3,072 tokens for MATH-500 and GSM8K, and 4,096 tokens for AIME24. Batch size is fixed per model (16 for LLaMA-3.1-8B-Instruct and Qwen2.5-Math-7B variants, 32 for Qwen2.5-Math-1.5B variants).

Table 6: Accuracy (%) across model configurations, injection positions, and benchmarks. Values in parentheses indicate the difference from the baseline. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Mode	MATH-500	GSM8K	AIME24
Instruct Models				
LLaMA-3.1-8B-Instruct	Baseline	**52.20**	85.75	6.67
	Prefix	45.00 (−7.2)	76.35 (−9.4)	3.33 (−3.3)
	Infix	49.40 (−2.8)	85.97 (+0.2)	**10.00** (+3.3)
	Suffix	49.40 (−2.8)	**86.73** (+1.0)	**10.00** (+3.3)
Qwen2.5-Math-7B-Instruct	Baseline	83.20	95.45	13.33
	Prefix	81.40 (−1.8)	94.92 (−0.5)	13.33 (+0.0)
	Infix	**84.20** (+1.0)	95.60 (+0.2)	**16.67** (+3.3)
	Suffix	83.40 (+0.2)	**95.68** (+0.2)	**16.67** (+3.3)
Qwen2.5-Math-1.5B-Instruct	Baseline	73.00	85.29	10.00
	Prefix	74.80 (+1.8)	85.29 (+0.0)	13.33 (+3.3)
	Infix	**75.20** (+2.2)	**85.75** (+0.5)	6.67 (−3.3)
	Suffix	74.20 (+1.2)	84.69 (−0.6)	**20.00** (+10.0)
Base Models (Raw Text)				
Qwen2.5-Math-7B	Baseline	**72.20**	84.15	6.67
	Prefix	71.60 (−0.6)	**84.31** (+0.2)	**16.67** (+10.0)
	Infix	63.20 (−9.0)	64.29 (−19.9)	**16.67** (+10.0)
	Suffix	55.60 (−16.6)	54.13 (−30.0)	13.33 (+6.7)
Qwen2.5-Math-1.5B	Baseline	61.40	**77.86**	**16.67**
	Prefix	**64.40** (+3.0)	74.30 (−3.6)	10.00 (−6.7)
	Infix	63.20 (+1.8)	73.92 (−3.9)	10.00 (−6.7)
	Suffix	54.60 (−6.8)	58.61 (−19.3)	3.33 (−13.3)
Base Models + ChatML (Format Mismatch)				
Qwen2.5-Math-7B	Baseline	52.20	58.30	**23.33**
	Prefix	59.20 (+7.0)	56.41 (−1.9)	3.33 (−20.0)
	Infix	62.00 (+9.8)	69.07 (+10.8)	**23.33** (+0.0)
	Suffix	**70.40** (+18.2)	**75.97** (+17.7)	**23.33** (+0.0)
Qwen2.5-Math-1.5B	Baseline	34.40	37.45	13.33
	Prefix	**63.40** (+29.0)	**67.02** (+29.6)	**20.00** (+6.7)
	Infix	39.20 (+4.8)	50.42 (+13.0)	6.67 (−6.7)
	Suffix	51.00 (+16.6)	50.64 (+13.2)	13.33 (+0.0)

Table 7: Number of RSP tokens (L) used for each model and benchmark.

Model	MATH-500	GSM8K	AIME24
LLaMA-3.1-8B-Instruct	15	20	15
Qwen2.5-Math-7B-Instruct	20	20	20
Qwen2.5-Math-1.5B-Instruct	20	10	20
Qwen2.5-Math-7B	10	10	20
Qwen2.5-Math-1.5B	10	10	20
Qwen2.5-Math-1.5B (ChatML)	20	20	10
Qwen2.5-Math-7B (ChatML)	20	20	20

A.1 RSP Injection Positions and Prompt Templates

Below we show the prompt structure for each injection position across all model types. Blue text indicates RSP embeddings, and purple text indicates special tokens.

A.1.1 Prefix

RSP embeddings are concatenated before the prompt embeddings. No text modification is applied.

ChatML (Qwen Instruct / Base + ChatML).

```
[Random Embeddings (L tokens)]
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
[Random Embeddings (L tokens)]
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
[Random Embeddings (L tokens)]
{question}
Please reason step by step, and put your final answer within \boxed{}.
```

A.1.2 Infix

A description text and special tokens are inserted into the prompt. The special token positions are then replaced with RSP embeddings at the embedding level.

ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens: <special_token_1>...<special_token_L>
<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
```

```
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens:
<reserved_special_token_0>...
<reserved_special_token_L>
<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens: <special_token_1>...<special_token_L>
Please reason step by step, and put your final answer within \boxed{}.
```

A.1.3 Suffix

For chat-templated models, RSP embeddings are inserted immediately before the end-of-turn token. For raw text models, RSP embeddings are concatenated at the end of the prompt.

ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}[Random Embeddings (L tokens)]<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}[Random Embeddings (L tokens)]<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
{question}
Please reason step by step, and put your final answer within \boxed{}.[Random
Embeddings (L tokens)]
```

A.2 Answer Extraction and Grading

The final answer is extracted from the model output through a fallback chain. Each step is attempted in order; if extraction fails, the next step is tried.

Answer Extraction Fallback Chain

1. `"final answer is $...$. I hope"` pattern (Minerva format)
2. `\boxed{...}`: last non-empty box is used; empty `\boxed{}` are skipped
3. Text following `"the answer is"`
4. Text following `"final answer is"`
5. Last number in the output (regex fallback)

Extracted answers are normalized by removing `$`, `\left`, `\right`, and unit strings. Grading is performed via symbolic equivalence checking: exact string match, numerical comparison (`math.isclose`, `rel_tol=1e-4`), and SymPy symbolic simplification.

A.3 Reproduced Prior Work Hyperparameters

All prior methods are reproduced on Qwen2.5-Math-1.5B-Instruct and Qwen2.5-Math-7B-Instruct, evaluated with greedy decoding (temperature $= 0$) and our unified answer extraction pipeline.

TTSV. Prefix length 20, trained for 20 epochs with AdamW optimizer, batch size 2, gradient accumulation 8 steps, and linear learning rate schedule. Learning rate is 1e-3 for Qwen-1.5B and 1e-5 for Qwen-7B.

LTPO. Table 8 reports the per-benchmark hyperparameters.

Table 8: LTPO hyperparameters.

Parameter	GSM8K	MATH-500	AIME24
tokens	8	8	8
steps	20	20	20
top_k	10	10	10
sigma	20	20	5
sigma_decay	0.95	0.95	0.95
lr	1e-2	5e-3	5e-2

SoftCoT. Projection module trained for 10 epochs. Thought tokens: 32 during training, 4 during evaluation. Batch size 4 for GSM8K, 1 for MATH with gradient accumulation 4. The small (assistant) model is Qwen2.5-Math-1.5B-Instruct in both configurations, while the large model is Qwen2.5-Math-1.5B-Instruct (learning rate 2e-5) or Qwen2.5-Math-7B-Instruct (learning rate 5e-6). For MATH-500 and AIME24 evaluation, we use the projection module trained on GSM8K, as it yields higher accuracy than the one trained on MATH.

A.4 Full Entropy Curves

Figure 5 shows the full normalized (0–100%) generation trajectories for entropy, top-1 probability, and varentropy. All methods converge to similar levels after the initial generation stage, confirming that the distributional changes induced by RSP are localized to the early reasoning phase.

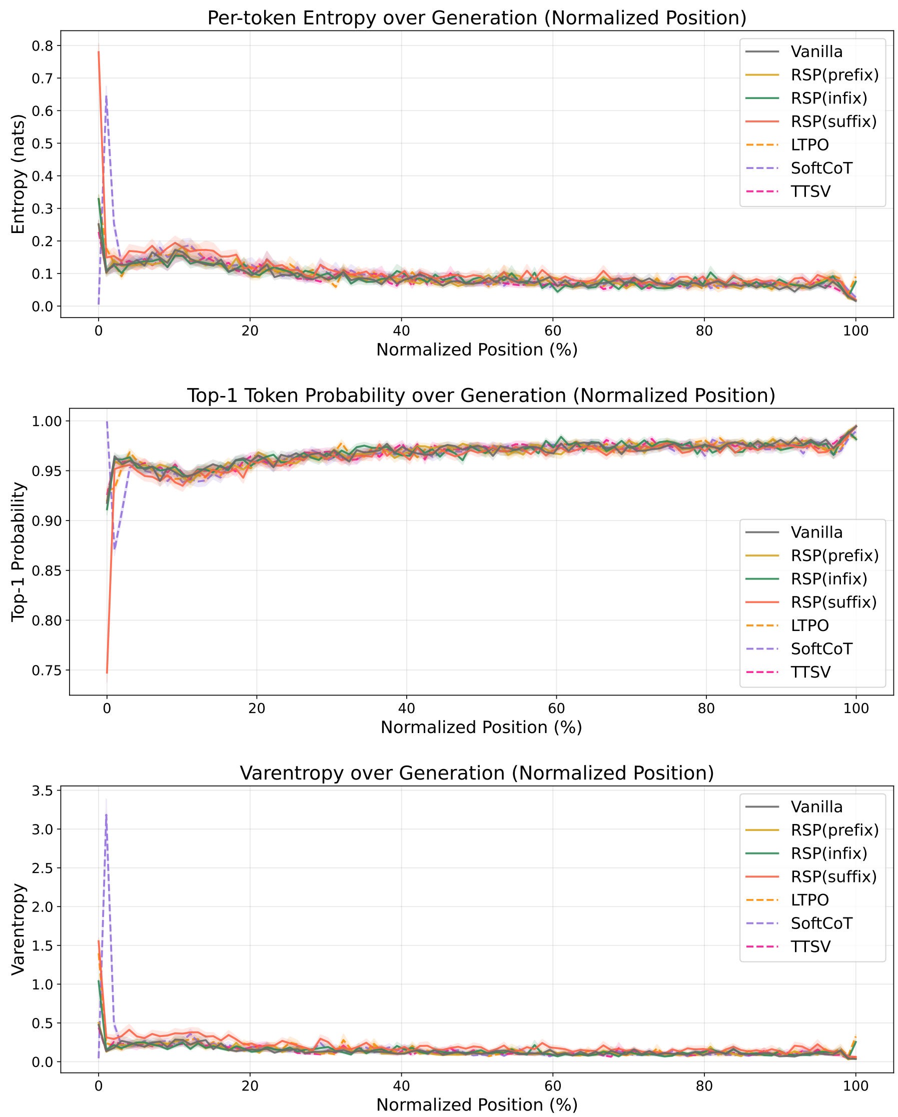


Figure 5: Mean entropy (top), top-1 probability (middle), and varentropy (bottom) over the full normalized generation trajectory (0–100%). Averaged over MATH-500, shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: existing soft prompt methods. All methods converge after the initial stage.

B Additional Semantic Diversity Analysis

This appendix complements the semantic diversity results reported in Section 3.4. We include per-problem t-SNE plots across a broader set of problems.

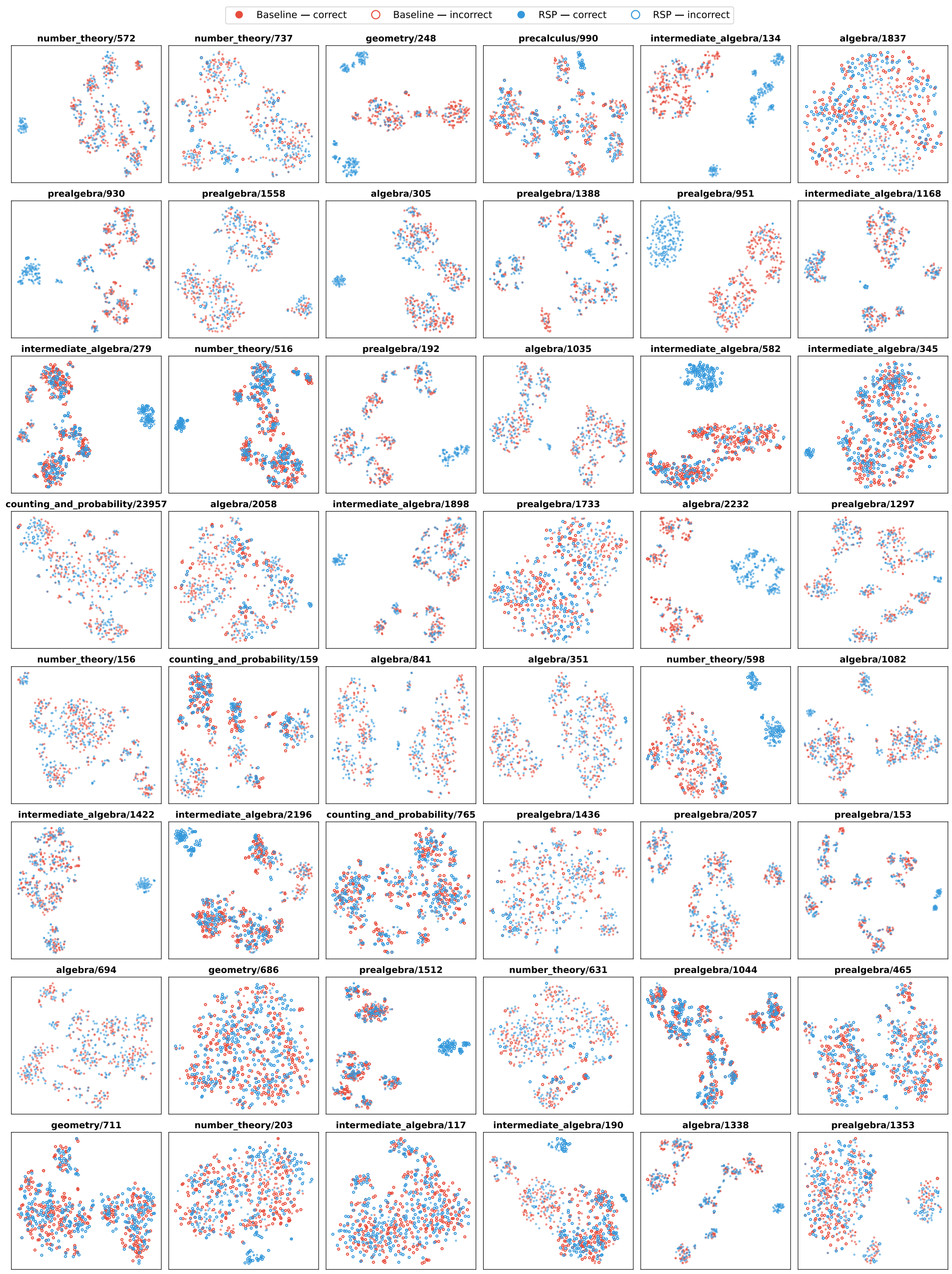


Figure 6: t-SNE projections of BGE-M3 embeddings for 48 out of the 64 evaluated MATH-500 problems (6×8 grid). Each subplot shows 300 baseline (red) and 300 RSP suffix (blue) samples under $\tau = 1.0$ with Qwen2.5-Math-1.5B-Instruct; filled markers denote correct answers and open markers denote incorrect. Subplot titles indicate `subject/problem_id`. Across most problems, RSP trajectories spread across multiple distinct regions while baseline samples concentrate in a single dominant cluster.

C Token-Level Distribution Metrics

This appendix provides formal definitions for the three metrics used in the Token-level analysis of Section 3.4. Let P_{base} denote the baseline next-token distribution at $\tau = 1.0$ and P_{target} the candidate distribution being compared (e.g., the baseline at $\tau = 2.0$ or RSP at $\tau = 1.0$). All metrics are computed analytically from saved logits at the first generation step.

Spearman rank correlation. Spearman's ρ is the Pearson correlation coefficient between the token ranks of two distributions:

$$\rho \;=\; \mathrm{Pearson}(\mathrm{rank}(P_{\text{target}}),\; \mathrm{rank}(P_{\text{base}}))\,. \tag{4}$$

Because temperature scaling $P^{(\tau)} \propto P^{1/\tau}$ is a strictly monotone transform, the rank order of tokens is preserved exactly, so $\rho = 1$ for any $\tau > 0$ as a mathematical identity. Any value $\rho < 1$ therefore indicates rank reorganization beyond what temperature scaling can produce.

Mass outside top-K. Let $\mathrm{TopK}(P_{\text{base}})$ be the set of K tokens with the highest probability under P_{base}. The probability mass that the candidate distribution places *outside* this set is

$$\mathrm{MassOutside}_K \;=\; 1 - \sum_{t \in \mathrm{TopK}(P_{\text{base}})} P_{\text{target}}(t). \tag{5}$$

This directly measures support expansion: how much probability the candidate assigns to tokens that are not in the baseline's preferred set. Reported values use $K = 10$.

Jensen–Shannon divergence. The symmetrized, bounded version of Kullback–Leibler divergence:

$$\mathrm{JS}(P, Q) \;=\; \tfrac{1}{2}\,\mathrm{KL}(P \,\|\, M) + \tfrac{1}{2}\,\mathrm{KL}(Q \,\|\, M), \qquad M = \tfrac{1}{2}(P + Q). \tag{6}$$

We use base-2 logarithms so that JS $\in [0, 1]$. Tokens absent from the stored top-100 are assigned a sentinel logit (-10^9) before softmax, which is negligible for our setting since the top-5 covers more than 99.9% of the probability mass.

D Why Random Prompts Are Semantically Neutral and Direction-Agnostic

This appendix formalizes the "why random" claim used in Section 4.1. The point is not that random prompts contain hidden task information. The point is structural: if the perturbation is meant to act as a generic exploration device rather than as a learned prompt prior, then it should be resampled across runs, centered so that it does not induce a fixed first-order drift, and direction-agnostic so that it does not under-serve whichever local branch direction matters for the current prompt.

D.1 Semantic Neutrality

Let $\bar{h} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_d)$ and $\hat{h} := \bar{h}/\|\bar{h}\|$ whenever $\bar{h} \neq 0$. Let $u_1, \ldots, u_M \in \mathbb{S}^{d-1}$ be any fixed unit vectors representing semantic anchor directions.

Lemma 2 (Semantic neutrality). *For any $\delta \in (0, 1)$,*

$$\Pr\left[\max_{r \in [M]} |\hat{h}^\top u_r| \leq \sqrt{\frac{2\log(2M/\delta)}{d-1}}\right] \geq 1 - \delta.$$

Hence the normalized direction of an isotropic Gaussian prompt is simultaneously nearly orthogonal to every fixed finite collection of semantic anchors with high probability.

Proof. Because $\bar{h}$ is isotropic Gaussian, its normalized direction $\hat{h}$ is uniformly distributed on the sphere $\mathbb{S}^{d-1}$. For a fixed unit vector $u \in \mathbb{S}^{d-1}$, the marginal $\hat{h}^\top u$ satisfies the standard spherical concentration bound [Vershynin, 2018, Theorem 3.4.6][Boucheron et al., 2013, Section 3.2],

$$\Pr\left(|\hat{h}^\top u| \geq t\right) \leq 2\exp\left(-\frac{(d-1)t^2}{2}\right) \qquad \text{for } t \in (0, 1),$$

which follows from the spherical-cap area bound $\sigma\{\hat{h} \in \mathbb{S}^{d-1} : \hat{h}^\top u \geq t\} \leq \frac{1}{2}(1 - t^2)^{(d-1)/2}$ together with $(1 - t^2)^{(d-1)/2} \leq \exp(-(d-1)t^2/2)$ derived from $\log(1 - x) \leq -x$ for $x \in [0, 1)$. Applying this bound to each u_r and using a union bound,

$$\Pr\left[\max_{r \in [M]} |\hat{h}^\top u_r| \geq t\right] \leq 2M \exp\left(-\frac{(d-1)t^2}{2}\right).$$

Setting

$$t = \sqrt{\frac{2\log(2M/\delta)}{d-1}}$$

gives the claim. If the displayed value of t exceeds 1, the statement is trivial because $|\hat{h}^\top u_r| \leq 1$ always. $\square$

The contrast with a token embedding is immediate. If $e_s \neq 0$ is a fixed token embedding and $\hat{e}_s := e_s/\|e_s\|$, then choosing the anchor direction $u = \hat{e}_s$ gives

$$|\hat{e}_s^\top u| = 1.$$

A token prompt is therefore maximally aligned with at least one semantic direction, whereas an isotropic random prompt is nearly orthogonal to every fixed finite set of such directions. This is the sense in which tokens inject semantic commitment while random prompts act as semantically uncommitted perturbations.

D.2 Centered Perturbations Do Not Impose Systematic First-Order Drift

Let Δ_{ij} be the baseline logit gap between tokens i and j, and write the first-order surrogate as

$$\Delta_{ij}(\bar{h}) = \Delta_{ij} + b_{ij}^\top \bar{h}.$$

Proposition 3 (No systematic first-order drift). *If $\mathbb{E}[\bar{h}] = 0$, then*

$$\mathbb{E}\big[\Delta_{ij}(\bar{h})\big] = \Delta_{ij}$$

for every token pair (i, j). If a deterministic prompt h_0 is reused across runs, then

$$\Delta_{ij}(h_0) - \Delta_{ij} = b_{ij}^\top h_0$$

is a fixed directional bias.

Proof. The centered case is immediate from linearity of expectation:

$$\mathbb{E}\big[\Delta_{ij}(\bar{h})\big] = \Delta_{ij} + b_{ij}^\top \mathbb{E}[\bar{h}] = \Delta_{ij}.$$

The deterministic case follows by substitution. □

The proposition does not say that a centered perturbation leaves every *realization* unchanged; it says there is no systematic first-order push toward one branch across runs. This is exactly the behavior desired from a generic exploration device. A fixed token or fixed soft prompt may still be useful, but it acts as a persistent prompt prior rather than as an unbiased exploration mechanism.

D.3 Direction-Agnostic Coverage

The second requirement is directional coverage. Let D be any zero-mean perturbation distribution in $\mathbb{R}^d$ with covariance Σ_D and variance budget $\mathrm{tr}(\Sigma_D) \le \rho^2$. For a unit vector b, the first-order perturbation energy available along b is $\mathrm{Var}_{h\sim D}(b^\top h) = b^\top \Sigma_D b$. This is the quantity that appears in the local linear logit surrogate used in the main text.

Theorem 3 (Direction-agnostic coverage)**.** *Define*

$$\mathcal{E}(D) := \min_{\|b\|=1} \mathrm{Var}_{h\sim D}(b^\top h).$$

Then

$$\mathcal{E}(D) = \lambda_{\min}(\Sigma_D) \le \frac{\rho^2}{d},$$

with equality if and only if $\Sigma_D = (\rho^2/d)\mathbf{I}_d$.

Proof. For any symmetric covariance matrix Σ_D,

$$\min_{\|b\|=1} b^\top \Sigma_D b = \lambda_{\min}(\Sigma_D),$$

so $\mathcal{E}(D) = \lambda_{\min}(\Sigma_D)$. Let the eigenvalues of Σ_D be $\lambda_1, \dots, \lambda_d$. Then

$$\lambda_{\min}(\Sigma_D) \le \frac{1}{d}\sum_{m=1}^{d} \lambda_m = \frac{\mathrm{tr}(\Sigma_D)}{d} \le \frac{\rho^2}{d}.$$

Equality holds if and only if all eigenvalues are equal, i.e., $\Sigma_D = (\rho^2/d)\mathbf{I}_d$. □

This theorem is a statement about second moments, not about Gaussianity by itself. Many distributions could realize isotropic covariance. We use a Gaussian in RSP because it is the simplest centered, rotation-symmetric family that matches the scale of pretrained embeddings. The theorem clarifies the structural advantage: a fixed prompt has $\Sigma_D = 0$ and therefore no across-run exploration, whereas any anisotropic perturbation leaves at least one branch-sensitive direction less explored than the isotropic optimum.

E Proofs for Attention-Mediated Annealing

E.1 Exact Attention Decomposition and Magnitude Control

Proposition 4 (Attention decomposition)**.** *Let $\alpha_{ij}^{(l)}$ be the attention weights for query position i at layer l, with n real tokens and k random tokens. Define*

$$w_{r,i}^{(l)} := \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)}$$

and, for $j \in [n]$,

$$\tilde{\alpha}_{ij}^{(l)} := \frac{\alpha_{ij}^{(l)}}{1 - w_{r,i}^{(l)}}.$$

Then

$$\mathbf{x}_i^{(l+1)} = (1 - w_{r,i}^{(l)})\, \tilde{\mathbf{x}}_i^{(l+1)} + \boldsymbol{\eta}_i^{(l)},$$

where

$$\tilde{\mathbf{x}}_i^{(l+1)} := \sum_{j=1}^{n} \tilde{\alpha}_{ij}^{(l)} \mathbf{v}_j^{(l)} \qquad \text{and} \qquad \boldsymbol{\eta}_i^{(l)} := \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)}.$$

Proof. The attention output is

$$\mathbf{x}_i^{(l+1)} = \sum_{j=1}^{n} \alpha_{ij}^{(l)} \mathbf{v}_j^{(l)} + \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)}.$$

By definition, $\sum_{j=1}^{n} \alpha_{ij}^{(l)} = 1 - w_{r,i}^{(l)}$, so

$$\sum_{j=1}^{n} \alpha_{ij}^{(l)} \mathbf{v}_j^{(l)} = (1 - w_{r,i}^{(l)}) \sum_{j=1}^{n} \frac{\alpha_{ij}^{(l)}}{1 - w_{r,i}^{(l)}} \mathbf{v}_j^{(l)} = (1 - w_{r,i}^{(l)})\, \tilde{\mathbf{x}}_i^{(l+1)}.$$

Substituting this identity gives the decomposition. □

Corollary 2 (Magnitude scales with random-token attention)**.** *For every realization of the random values,*

$$\|\boldsymbol{\eta}_i^{(l)}\| \le w_{r,i}^{(l)} \max_{j \in [k]} \|\mathbf{v}_{r_j}^{(l)}\|.$$

Proof. By the triangle inequality,

$$\|\boldsymbol{\eta}_i^{(l)}\| = \left\| \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)} \right\| \le \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)} \|\mathbf{v}_{r_j}^{(l)}\| \le w_{r,i}^{(l)} \max_{j \in [k]} \|\mathbf{v}_{r_j}^{(l)}\|.$$

□

Corollary 2 is the only quantitative fact needed in the main text. No independence assumption between attention weights and random keys is required. Once $w_{r,i}^{(l)}$ is small, the random contribution must be small as well.

E.2 Attention Decay Bound

Theorem 4 (Attention decay)**.** *The total attention on random tokens satisfies*

$$w_{r,i}^{(l)} \le \frac{k}{k + n \cdot \exp(\Delta_i^{(l)}/\sqrt{d})}, \tag{7}$$

where $\Delta_i^{(l)} := \min_{j \in [n]} \mathbf{q}_i^\top \mathbf{k}_j - \max_{j' \in [k]} \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}$, is the real-vs-random logit gap.

Proof. Let $s_j = \mathbf{q}_i^\top \mathbf{k}_j/\sqrt{d}$ for real keys and $s_{r_{j'}} = \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}/\sqrt{d}$ for random keys. By definition of $\Delta_i^{(l)}$, each real key satisfies $s_j \geq \Delta_i^{(l)}/\sqrt{d} + s_{r,\max}$ where $s_{r,\max} = \max_{j'} s_{r_{j'}}$. Therefore $\exp(s_j) \geq \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot \exp(s_{r,\max})$ for every real key. Summing over n real keys gives

$$\sum_{j=1}^{n} \exp(s_j) \geq n \cdot \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot \exp(s_{r,\max}) \geq n \cdot \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot \frac{1}{k} \sum_{j'=1}^{k} \exp(s_{r_{j'}}).$$

Let $R := \sum_{j'=1}^{k} \exp(s_{r_{j'}})$. Then

$$w_{r,i}^{(l)} = \frac{R}{\sum_{j=1}^{n} \exp(s_j) + R} \leq \frac{R}{n \cdot \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot R/k + R} = \frac{k}{k + n \cdot \exp(\Delta_i^{(l)}/\sqrt{d})}.$$

□

Corollary 3 (Sequence-wise annealing under a fixed logit gap)**.** *For fixed k and $\Delta_i^{(l)}$, the upper bound*

$$f(n) = \frac{k}{k + n \exp(\Delta_i^{(l)}/\sqrt{d})}$$

is strictly decreasing in n. Consequently, cache growth alone decreases the maximum perturbation mass compatible with that gap.

Proof. Differentiate $f(n)$ with respect to n:

$$f'(n) = -\frac{k \exp(\Delta_i^{(l)}/\sqrt{d})}{\left(k + n \exp(\Delta_i^{(l)}/\sqrt{d})\right)^2} < 0.$$

Thus $f(n)$ is strictly decreasing. □

Corollary 4 (Vanishing random contribution under annealing)**.** *If $n \to \infty$ along decoding while k and $\Delta_i^{(l)}$ remain fixed, then $w_{r,i}^{(l)} \to 0$. Moreover, for any fixed realization of the k random values,*

$$\|\boldsymbol{\eta}_i^{(l)}\| \leq w_{r,i}^{(l)} \max_{j \in [k]} \|\mathbf{v}_{r_j}^{(l)}\| \to 0.$$

If the random values are Gaussian, the displayed maximum is almost surely finite because k is finite.

Proof. Corollary 3 implies $w_{r,i}^{(l)} \to 0$. The bound on $\|\boldsymbol{\eta}_i^{(l)}\|$ is Corollary 2. Since the multiplier $\max_{j \in [k]} \|\mathbf{v}_{r_j}^{(l)}\|$ is finite for any fixed realization of finitely many sampled values, the product converges to zero. □

E.3 Proofs for Output Distribution Shift

The main text makes two output-level claims. Proposition 1 establishes pairwise rank randomness, and Proposition 2 upgrades that statement to top-K entry. Both are statements about a local first-order surrogate of the perturbed logits.

E.3.1 Autoregressive Formulation

In autoregressive decoding, the joint distribution over a sequence $y_{1:T}$ factorizes as

$$\Pr(y_{1:T} \mid x) = \prod_{t=1}^{T} p(y_t \mid x, y_{<t}),$$

where each step is governed by a prefix-conditioned distribution. Any perturbation introduced by RSP affects generation through a sequence of local changes to $p(y_t \mid x, y_{<t})$, rather than a single global distribution.

E.3.2 Local Linearization of Logits under RSP

Let $\bar{h}$ denote the centered random soft prompt. We model the perturbed logits as a function $z_i(\bar{h})$, assuming local smoothness in a neighborhood of $\bar{h} = 0$:

$$z_i(\bar{h}) = z_i(0) + \nabla_h z_i(0)^\top \bar{h} + r_i(\bar{h}),$$

where $r_i(\bar{h}) = O(\|\bar{h}\|^2)$. Defining $a_i := \nabla_h z_i(0)$, we obtain the local linear approximation $z_i(\bar{h}) \approx z_i + a_i^\top \bar{h}$. This approximation is valid for sufficiently small $\|\bar{h}\|$ and is used as a first-order surrogate for the direction of perturbation rather than as an exact global model of the transformer logits.

E.3.3 Proof of Proposition 1 (Stochastic Ranking)

If there exists a token pair (i, j) such that $a_i \neq a_j$, then $0 < \Pr_{\bar{h}}(z_i(\bar{h}) > z_j(\bar{h})) < 1$.

Proof. Under the local linear approximation,

$$z_i(\bar{h}) - z_j(\bar{h}) = (z_i - z_j) + (a_i - a_j)^\top \bar{h}.$$

Let $b_{ij} := a_i - a_j \neq 0$. If $\bar{h} \sim \mathcal{N}(0, \sigma^2 I)$, then $b_{ij}^\top \bar{h} \sim \mathcal{N}(0, \sigma^2 \|b_{ij}\|^2)$, a non-degenerate Gaussian. The perturbed logit difference

$$\Delta_{ij}(\bar{h}) := (z_i - z_j) + b_{ij}^\top \bar{h}$$

therefore has positive probability mass on both sides of zero, which yields

$$0 < \Pr_{\bar{h}}(\Delta_{ij}(\bar{h}) > 0) < 1.$$

□

E.3.4 Proof of Proposition 2 (Top-K Entry)

Let $J_K(s_t)$ be the baseline top-K set at state s_t, and let $i \notin J_K(s_t)$. Define

$$\mathcal{G}_{i,K} := \left\{ \bar{h} \in \mathbb{R}^d : \#\left\{ j \neq i : z_j + a_j^\top \bar{h} > z_i + a_i^\top \bar{h} \right\} \leq K - 1 \right\}.$$

If $\mathcal{G}_{i,K}$ contains a nonempty open set and $\bar{h}$ is sampled from an isotropic Gaussian, then

$$\Pr_{\bar{h}} \left(i \in \mathrm{TopK}(p^{\mathrm{RSP}}(\cdot \mid s_t, \bar{h})) \right) > 0.$$

Proof. Under the local linear model, token i belongs to the perturbed top-K set if and only if at most $K - 1$ tokens outrank it, i.e., if and only if $\bar{h} \in \mathcal{G}_{i,K}$. For each subset S of token indices with $i \notin S$ and $|S| \leq K - 1$, define

$$\mathcal{O}_S := \left\{ \bar{h} : z_j + a_j^\top \bar{h} > z_i + a_i^\top \bar{h} \text{ for all } j \in S,\ z_i + a_i^\top \bar{h} > z_j + a_j^\top \bar{h} \text{ for all } j \notin S \cup \{i\} \right\}.$$

Each $\mathcal{O}_S$ is an intersection of open half-spaces and is therefore open. Moreover,

$$\mathcal{G}_{i,K} = \bigcup_{S:\, i \notin S,\ |S| \leq K-1} \mathcal{O}_S,$$

so $\mathcal{G}_{i,K}$ is a finite union of open sets and is itself open. By assumption it contains a nonempty open set. The isotropic Gaussian has a density that is strictly positive everywhere on $\mathbb{R}^d$, so every nonempty open set has positive probability measure. Therefore

$$\Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0,$$

which is exactly the event that token i enters the top-K set under the perturbed distribution. □

E.3.5 Trajectory-Level Implications

Since autoregressive generation composes conditional distributions across steps, perturbations at early steps can lead to distinct trajectories. Let $\mathcal{R}_{k,K}(x)$ denote the set of trajectories containing at least one token outside the baseline top-K set. If RSP increases the accessibility of such tokens at early steps, then the probability of sampling trajectories in $\mathcal{R}_{k,K}(x)$ increases under repeated sampling. The key point is temporal compounding: one early top-K entry changes the next decoding state, which changes the next candidate set, and so on. This is the formal bridge from local rank perturbations to global trajectory diversity.

F Attention Mass Analysis

This appendix formalizes the per-token attention mass reported in Figure 4 and the exclusion criteria applied to the reasoning span and the layer axis. For each of the 500 MATH-500 problems, a fresh RSP $\mathbf{H} \in \mathbb{R}^{L \times d}$ with $L = 10$ is sampled independently, so the reported heatmaps average over both problem variability and RSP-vector variability.

F.1 Per-Token Attention Mass

For each problem, we measure attention on the concatenated sequence [prompt; $\mathbf{H}$; generation], where the generation is the trajectory produced by the same model under matching RSP injection. Let $\alpha_{i,j}^{(\ell,h)}$ denote the post-softmax attention weight at layer ℓ, head h, from query position i to key position j, satisfying $\sum_j \alpha_{i,j}^{(\ell,h)} = 1$ under the causal mask. Let Q and R denote the index sets of question and RSP tokens with sizes $|Q|$ and $|R| = L$, and let H be the number of heads. The per-token attention mass that query i directs to each group at layer ℓ is

$$\mathrm{PTM}_Q[\ell, i] = \frac{1}{|Q|H} \sum_{h=1}^{H} \sum_{j \in Q} \alpha_{i,j}^{(\ell,h)}, \qquad \mathrm{PTM}_R[\ell, i] = \frac{1}{|R|H} \sum_{h=1}^{H} \sum_{j \in R} \alpha_{i,j}^{(\ell,h)}. \tag{8}$$

Normalization by $|Q|$ and $|R|$ controls for group size, so both quantities express how much attention a single question (resp. RSP) token receives on average under the same softmax denominator. Question tokens thereby serve as a size-matched baseline that absorbs sequence-length effects common to both groups.

F.2 Heatmap Aggregation

We partition the layer indices and the reasoning position indices each into five equal-size bins $\{B_L^{(b)}\}_{b=1}^5$ and $\{B_P^{(b)}\}_{b=1}^5$. For sample s and target group $\bullet \in \{Q, R\}$, the value in cell (b_L, b_P) is

$$M_\bullet^{(s)}[b_L, b_P] = \frac{1}{|B_L^{(b_L)}| \cdot |B_P^{(b_P)}|} \sum_{\ell \in B_L^{(b_L)}} \sum_{i \in B_P^{(b_P)}} \mathrm{PTM}_\bullet[\ell, i]. \tag{9}$$

Figure 4 reports the sample average of Eq. (9) over the 500 valid samples.

F.3 Excluding the `\boxed{...}` Span

The reasoning position range terminates strictly before the final `\boxed{...}` span. Chain-of-thought outputs decompose into a *text* (content) component and a *pattern* (template) component that contribute independently to downstream performance [Madaan and Yazdanbakhsh, 2022]. The `\boxed{...}` wrapper is a canonical pattern: a short, stereotyped span whose role is to complete the answer format rather than to extend reasoning. Including it would therefore inject a template-driven attention signature unrelated to reasoning dynamics.

F.4 Excluding the Final Two Layers

The layer axis further excludes the last two transformer layers. Two well-documented properties of late layers, none specific to our setup, motivate this choice.

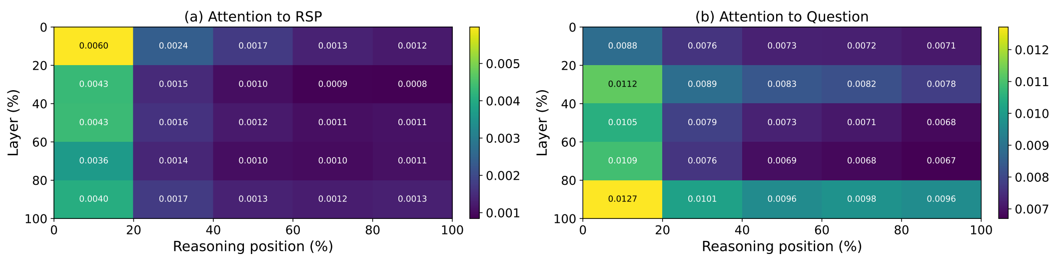
Output-projection specialization. Late-layer hidden states lie in an approximately linear relationship with vocabulary logits and therefore function as a progressive linear readout rather than as representation-transforming blocks [Belrose et al., 2023]. Consistently, late-layer feed-forward modules have been characterized as output-distribution shaping mechanisms [Geva et al., 2021]. Attention in these layers therefore reflects output formation rather than reasoning computation.

Register-token hypothesis. Uninformative patch tokens in Vision Transformers have been shown to be repurposed as *registers*: storage slots that absorb attention in deeper layers [Darcet et al., 2024]. Because RSPs share a similar profile as continuous tokens without pretrained semantic content, a similar repurposing in late transformer layers cannot be ruled out. Excluding the final two layers removes this potential register-induced confound from the reasoning-phase signal.

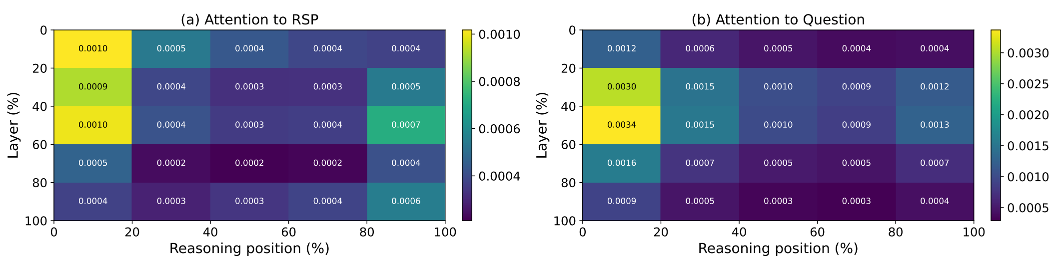
The two lines of evidence above identify known late-layer phenomena — linear readout and potential register repurposing — whose attention signatures are orthogonal to reasoning-phase dynamics. Removing the final two layers preempts their contamination of the reasoning-phase signal, yielding a principled, literature-grounded exclusion rather than a model-specific tuning decision.

F.5 Additional Suffix Heatmaps

Figure 7 reports the same per-token RSP attention analysis under suffix injection for Qwen2.5-Math-1.5B-Instruct and LLaMA-3.1-8B-Instruct. For the late-layer reasons discussed above, the pattern does not generalize uniformly across every cell; nevertheless, the overall trend is consistent across both models: question-token attention varies broadly across layers, whereas RSP attention shows the layer-wise and sequence-wise decay predicted by Eq. (3).



(a) Qwen2.5-Math-1.5B-Instruct, suffix



(b) LLaMA-3.1-8B-Instruct, suffix

Figure 7: Per-token RSP attention mass under suffix injection for the remaining two models (500 MATH-500 samples each). Axes and preprocessing match Figure 4: x-axis is reasoning position binned into five quantiles, y-axis is layer depth binned into five quantiles (top = shallow), color encodes the 500-sample mean per-token attention assigned to RSP tokens, and tokens inside `\boxed{}` spans and the last two layers are excluded.

G GRPO Formulation

For a given prompt q, the behavior policy $\pi_{\theta_{\text{old}}}$ samples a group of G responses $\{o_i\}_{i=1}^{G}$ with corresponding rewards $\{R_i\}_{i=1}^{G}$. The advantage of each response is computed by normalizing the group-level rewards:

$$\hat{A}_{i,t} = \frac{R_i - \text{mean}(\{R_i\}_{i=1}^{G})}{\text{std}(\{R_i\}_{i=1}^{G})}. \tag{10}$$

The policy is updated via a clipped objective with a KL penalty:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\Big(\min\Big(r_{i,t}\hat{A}_{i,t},\ \text{clip}(r_{i,t}, 1-\varepsilon, 1+\varepsilon)\,\hat{A}_{i,t}\Big) - \beta D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})\Big)\right], \tag{11}$$

where $r_{i,t} = \pi_\theta(o_{i,t} \mid q, o_{i,<t})/\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})$ is the importance sampling ratio, ε is the clipping range, and β controls the strength of the KL penalty against the reference policy π_{ref}.

H Broader Impacts

This work provides empirical and theoretical analysis of how random embedding injection affects the reasoning behavior of large language models. The contributions are primarily foundational, with potential downstream implications for reinforcement learning-based post-training of reasoning models such as GRPO.

Positive impacts. RSP-induced trajectory diversity could improve the sample efficiency of RL-based LLM post-training by providing richer learning signals within group-relative advantage estimation. This has the potential to reduce compute costs associated with alignment and reasoning-oriented fine-tuning, making such methods more accessible to research groups with limited resources.

Potential concerns. Methods that expand the effective output support of LLMs may increase the reachability of low-probability tokens, which includes both desirable alternative reasoning paths and potentially unsafe or off-distribution outputs. Practitioners applying RSP in production systems should combine it with existing safety filtering and alignment mechanisms rather than treating it as a standalone diversity source.

Limitations on impact scope. Since this work focuses on analysis rather than deploying a new model or dataset, the direct societal impact is limited. The broader implications described above are contingent on future work integrating RSP into applied training pipelines.

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